

Mean Field Games - Intro

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These notes discuss some motivation for studying Mean Field Games, introduce a few mathematical prerequisites, and then introduce the problem as described in Bensoussan, Frehse, and Yam “*Mean Field Games and Mean Field Type Control Theory*” Chapters 1, 2, and 3.1.

Motivation

- Heterogeneous agents + continuous time
 - Income and wealth inequality (Bewley-Huggett-Aiyagari)
 - Firm dynamics (Hopenhayn)
 - Demography and intergenerational risk (OLG)
 - Business cycles (Krusell-Smith, HANK)
- Mathematical foundations
 - Derive model formulas in general case
 - Alternative representations: State space (HJB-KFE) or sequence space (Maximum principle)
 - Establish properties: existence, uniqueness, regularity, quality of mean field approximation
 - Enable computation/analysis
- Extensions w/ new applications
 - Planner’s problem $\rightarrow$ Mean field control
 - Ramsay taxation $\rightarrow$ Stackelberg MFG
 - IO/Micro $\rightarrow$ MFG with strategic agents

Math prerequisites

I will skip these during lecture 1 due to time constraints and predicted participant background, down to Itô’s lemma and adjoints, unless asked. But I’m including a brief overview of prerequisites to establish notation and remind us older people of some useful facts.

Brownian Motion

A (standard) Brownian Motion/Wiener process in n dimensions on $[0, T]$ is a stochastic process (random function) $w(t)$ s.t.

1. $w(0) = 0$ (almost surely)
2. $w(t+u) - w(t)$, $u > 0$ is independent of $w(s)$ $\forall s < t$
3. $w(t+u) - w(t) \sim N(0, u \cdot I_n)$ $\forall t, \forall u > 0$
4. The paths, $w(\cdot) : [0, T] \rightarrow \mathbb{R}^n$ are (almost surely) continuous

One can show, with great effort, a process $w(t)$ with these properties exists and is not differentiable but is Hölder with exponent α for any $\alpha \in [0, 0.5)$.

If you know what a Σ -field is, there exists a probability space Ω and a *filtration* $\mathcal{F}_t$, a nested sequence of Σ -fields on Ω such that $\mathcal{F}_s \subseteq \mathcal{F}_t$ $\forall s < t$ such that $w(\cdot)$ is *adapted*: each $w(t)$ is $\mathcal{F}_t$ -measurable. If you don't know, replace " $\mathcal{F}_t$ -measurable" with "is a function of info up to time t " and otherwise ignore it.

A Wiener process arises naturally as the limit of any process summing up many small independent increments over time via a (Functional) Central Limit Theorem.

If $\{x_n\}_{n=1}^N$ are i.i.d. mean 0, variance 1 and $h \in [0, 1]$

$$\frac{1}{\sqrt{N}} \sum_{n=1}^{\lfloor hN \rfloor} x_n \xrightarrow{d} w(h)$$

where convergence in distribution is of the whole function $w(\cdot) : [0, 1] \rightarrow \mathbb{R}$ (in the Skorohod topology of càdlàg functions, or just uniform if you are willing to ignore some measurability problems that are addressed by the Hoffman-Jørgensen theory of convergence in outer measure).

Diffusion Processes

To model processes that have predictable dynamics over time, we use diffusion processes, where mean and variance to be a function of the current process value.

Let $\mu(\cdot, \cdot) : \mathbb{R}^n \times [0, T] \rightarrow \mathbb{R}^n$, $\sigma : \mathbb{R}^n \times [0, T] \rightarrow \mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)$ and $w(\cdot)$ be an n -dimensional standard Wiener process, then we can define a diffusion process, Itô process, or stochastic differential equation (SDE)

$$dX(t) = \mu(X(t), t)dt + \sigma(X(t), t)dw(t)$$

with initial condition $X(0) = x_0$

$$\text{Equivalently, } X(t) = x_0 + \int_0^t \mu(X(s), s)ds + \int_0^t \sigma(X(s), s)dw(s)$$

where the latter integral is an Itô integral, which satisfies, for a sequence of partitions $\pi_n := \{[t_{i-1}, t_i] : i = 1 \dots n\}$ of $[0, T]$ with $\lim_{n \rightarrow \infty} \sup_{[t_{i-1}, t_i] \in \pi_n} |t_i - t_{i-1}| \rightarrow 0$ and for any process $g(t)$ which is càdlàg (right-continuous with left limits), $\mathcal{F}_t$ adapted, and $\int_0^t g^2(s)ds < \infty$

$$\int_0^t g(s)dw(s) = \lim_{n \rightarrow \infty} \sum_{[t_{i-1}, t_i] \in \pi_n} g(t_{i-1})[w(t_i) - w(t_{i-1})].$$

This process is well-defined (almost surely exists, is unique, and is square integrable) under some standard regularity conditions on μ, σ , viz, for some $C, D < \infty$

$$|\mu(x, t)| + \|\sigma(x, t)\| < C(1 + |x|)$$

$$|\mu(x, t) - \mu(y, t)| + \|\sigma(x, t) - \sigma(y, t)\| < D(|x - y|)$$

where $\|\sigma\| = \sqrt{\text{tr}(\sigma\sigma^*)} = \sqrt{\sum_{i,j=1}^n |\sigma_{i,j}|^2}$ is the Frobenius norm on matrices.

Itô calculus

The result we will use most often is a version of the chain rule for diffusion processes. Let $f(x, t) : \mathbb{R}^n \times [0, T] \rightarrow \mathbb{R}$ be twice continuously differentiable and $X(t)$ be a diffusion process as above. Then if $Y(t) = f(X(t), t)$

$$dY(t) = \left[\frac{\partial}{\partial t} f(X(t), t) + Df(X(t), t)\mu(X(t), t) + \frac{1}{2}\text{tr}(\sigma(X(t), t)\sigma(X(t), t)^* D^2 f(X(t), t)) \right] dt + Df(X(t), t)\sigma(X(t), t)dw(t)$$

where Df is the x derivative of f in $\mathbb{R}^n \times [0, T] \rightarrow \mathcal{L}(\mathbb{R}^n, \mathbb{R}^1)$ and $D^2 f$ is the Hessian, in $\mathbb{R}^n \times [0, T] \rightarrow \mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)$

This follows from a second order Taylor expansion plus the result that a 1d Brownian motion $w(t)$ has quadratic variation $[w()]_t := \lim_{n \rightarrow \infty} \sum_{[t_{i-1}, t_i] \in \pi_n} (w(t_i) - w(t_{i-1}))^2 = t$ where $\pi(n)$ are partitions over $[0, t]$, by independent normal increments, so that, as a result, $\Delta w \Delta w^* = I_n \Delta t + o(\Delta t)$, and so $(dw(t))^2 = dt$

Infinitesimal generators and adjoints

It will be convenient to represent the change over time in a function of a diffusion process by an operator called an infinitesimal generator

Given an Itô process $dX(t) = \mu(X(t), t)dt + \sigma(X(t), t)dw(t)$, its *infinitesimal generator* is an operator $\mathcal{A}_t[\phi]$ that takes any function $\phi(\cdot, \cdot) \in \mathcal{C}^2(\mathbb{R}^d \times [0, T])$ and returns the time derivative of the expectation of $\phi(\cdot, \cdot)$ evaluated at some x, t

$$\frac{d}{ds} \Big|_{s=0} E[\phi(X(t+s), t+s) | X(t) = x] = \lim_{\Delta t \rightarrow 0} E\left[\frac{\phi(X(t+\Delta t), t+\Delta t) - \phi(X(t), t)}{\Delta t} | X(t) = x \right] = \frac{E[d\phi(X(t), t) | X(t)=x]}{dt}$$

Apply Itô's lemma:

$$= \frac{E[\frac{\partial}{\partial t}\phi(X(t),t)+D\phi(X(t),t)\mu(X(t),t)+\frac{1}{2}\text{tr}(\sigma(X(t),t)\sigma(X(t),t)^*D^2\phi(X(t),t))]dt+D\phi(X(t),t)\sigma(X(t),t)dw(t)]}{dt}$$

Use the fact that $E[f(x)dw(t)] = 0$ for any $f(\cdot)$ (ie, the diffusion part is a “semi-martingale”) to drop the last part, then cancel out the dt to obtain

$$\mathcal{A}_t[\phi](x) = \frac{\partial}{\partial t}\phi(x, t) + D\phi(x, t)\mu(x, t) + \frac{1}{2}\text{tr}(\sigma(x, t)\sigma(x, t)^*D^2\phi(x, t))$$

We will use this a lot.

Another formula we will use a lot is the definition of an L^2 adjoint. Given an inner product $\langle \cdot, \cdot \rangle$ and inner product space $\mathcal{H}$, and an operator $A : \mathcal{H} \rightarrow \mathcal{H}$, the *adjoint* of A , denoted $A^* : \mathcal{H} \rightarrow \mathcal{H}$, is defined as

$$\langle f, Ag \rangle = \langle A^*f, g \rangle \text{ for any two elements } f, g \text{ on } \mathcal{H}.$$

For $\mathcal{H} = \mathbb{R}^n$ with the Euclidean inner product, the adjoint is the transpose of an $n \times n$ matrix. On $\mathbb{C}^n$ with the standard inner product, it is the conjugate transpose. For differential operators on L^2 , it can be calculated by integration by parts.

Applying this to $\mathcal{A}$, we get that $\mathcal{A}^*[\phi(\cdot)](\cdot)$ satisfies

$$\int \psi(x)\mathcal{A}\phi(x)dx = \int \mathcal{A}^*\psi(x)\phi(x)dx \text{ for any } \phi, \psi \text{ in a suitable space of twice differentiable functions.}$$

Letting $a(x) = \frac{1}{2}\sigma(x)\sigma^*(x)$, by Itô’s lemma, the driftless diffusion x with noise $\sigma(x)$ applied to $\varphi(x)$ has infinitesimal generator $A\varphi(x) = -\text{tr } a(x)D^2\varphi(x)$

Applying the adjoint formula

$$\int A^*\psi(x)\varphi(x)dx = \int \psi(x)A\varphi(x)dx = -\int \psi(x)\text{tr } a(x)D^2\varphi(x)dx$$

Apply integration by parts to obtain

$$= -\int \sum_{k,l=1}^n \frac{\partial^2}{\partial x_k \partial x_l} (a_{kl}(x)\psi(x))\varphi(x)dx$$

$$\text{so } A^*\phi(x) = -\sum_{k,l=1}^n \frac{\partial^2}{\partial x_k \partial x_l} (a_{kl}(x)\phi(x))$$

For the drift part only, the infinitesimal generator is $M[\phi](x) = D\varphi(x)g(x)$

$$\int M^*\psi(x)\varphi(x)dx = \int \psi(x)M\varphi(x)dx = \int \psi(x)D\varphi(x)g(x)dx$$

integrate by parts to get

$$= \int \text{div}(g(x)\psi(x))\varphi(x)dx$$

$$\text{so } M^*\phi(x) = \text{div}(g(x)\phi(x))$$

For the time component, $H\phi(t) = \frac{\partial}{\partial t}\phi(t)$

$$\int_0^T \psi(t) \frac{\partial}{\partial t} \phi(t) dt = [\psi(T)\phi(T) - \psi(0)\phi(0)] - \int_0^T \frac{\partial}{\partial t} \psi(t) \phi(t) dt$$

So $H^*\phi(t) = \phi(t)\delta_{t=T} - \phi(t)\delta_{t=0} - \frac{\partial}{\partial t}\phi(t)$.

Definition and formulas

Problem Statement

We consider a population of a large number of agents, which we will take to be a continuum. Each agent faces a time-varying state $x(t) \in \mathbb{R}^n$ and makes decisions $\nu(t) \in \mathbb{R}^d$ over a period $[0, T]$ in order to optimize a payoff $f(t)$ in each instant and a terminal payoff $h(T)$ at T .

In a typical example, x may be wealth, ν consumption spending and portfolio investments, and the payoff dependent on the path of consumption and terminal wealth.

What makes it a game is that other agents' states matter for each individual agent's state and payoffs. E.g., spending of all agents may determine asset prices, which affect change in wealth and terminal wealth.

What makes it a “mean field” game is that to simplify, we assume that there are so many agents, each acting symmetrically and anonymously, that only the distribution of $x(t)$, denoted $m(t)$ rather than the individual values, matters to other agents. For now we will assume $m(t)$ may be represented by a density, as a function in $L^1(\mathbb{R}^d)$.

We will assume that each agent's state $x(t)$ follows a diffusion process

$$dx(t) = g(x(t), m(t), \nu(t))dt + \sigma(x(t))dw(t)$$

$$x(0) = x_0$$

where $g(x, m, \nu) : \mathbb{R}^n \times L^1(\mathbb{R}^n) \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ $\sigma(x) : \mathbb{R}^n \rightarrow \mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)$ satisfy the right regularity conditions, viz, differentiability and linear growth in x, ν , and each agent's $w(t)$ is independent of all other agents'.

Each agent acts rationally by choosing their actions ν by a feedback control $\nu(x(t))$ to minimize an objective

$$J(\nu(\cdot), m(\cdot)) = E \left[\int_0^T f(x(t), m(t), \nu(x(t)))dt + h(x(T), m(T)) \right]$$

where $f(x, m, \nu) : \mathbb{R}^n \times L^1(\mathbb{R}^n) \times \mathbb{R}^d \rightarrow \mathbb{R}$ $h(x, m) : \mathbb{R}^n \times L^1(\mathbb{R}^n) \rightarrow \mathbb{R}$ are differentiable and have quadratic growth in x, ν .

While each process is identical, the realizations will differ, inducing a time path of (generically) non-degenerate distributions $m(x, t)$, which we will assume is initialized at $m(x, 0) = m_0(x)$. In a Mean Field Game, agents optimize as if the process $m(t)$ is fixed, but their joint behavior determines the process.

Putting the setup together, an equilibrium of a Mean Field Game is a pair $\hat{\nu}(\cdot), m(\cdot)$ s.t. $\hat{x}(\cdot)$ is the solution of

$$d\hat{x}(t) = g(\hat{x}(t), m(t), \hat{\nu}(\hat{x}(t)))dt + \sigma(\hat{x}(t))dw(t)$$

$$\hat{x}(0) = x_0$$

$m(t)$ is the probability distribution of $\hat{x}(t) \forall t \in [0, T]$

and $J(\hat{\nu}(\cdot), m(\cdot)) \leq J(\nu(\cdot), m(\cdot))$

Distributional Dynamics - Kolmogorov Forward Equation

To figure out what is the probability distribution induced by a diffusion process $x(t)$, we can characterize its law of motion by a Fokker-Planck equation (also called a Kolmogorov Forward Equation).

Letting $a(x) = \frac{1}{2}\sigma(x)\sigma^*(x)$

by Itô's lemma, the driftless diffusion x with noise $\sigma(x)$ applied to $\varphi(x)$ has infinitesimal generator $A\varphi(x) = -\text{tra}(x)D^2\varphi(x)$

Adding drift $g(x)$, the action of the process is given by $\mathcal{A}\varphi(x) := D\varphi(x)g(x) + A\varphi(x)$.

To derive what happens to the distribution $m(t)$ of $x(t)$ over time, which is both the distribution for a single agent and the cross-sectional distribution by independence, consider taking the cross-sectional expectation at time t of any function $\varphi(\cdot)$

$$E[\varphi(x(t))] = \int \varphi(x)m(x, t)dx$$

take derivatives with respect to time, and one gets

$$\int \varphi(x) \frac{\partial}{\partial t} m(x, t) dx$$

Taking the distribution over time perspective, $\int \varphi(x)m(x, t+s)dx = E[\varphi(x(t+s))|x(0) = x_0]$ also since $m(x, t)$ is the distribution of a single agent. By the Law of Iterated Expectations, this is

$$E[E[\varphi(x(t+s))|x(t) = x(t)]|x(0) = x_0] = \int E[\varphi(x(t+s))|x(t) = x]m(x, t)dx$$

take time derivatives to get $\int \frac{d}{ds}|_{s=0} E[\varphi(x(t+s))|x(t) = x]m(x, t)dx = \int \mathcal{A}\varphi(x)m(x, t)dx$ on the RHS and on the LHS

$$\frac{d}{ds}|_{s=0} \int \varphi(x)m(x, t+s)dx = \int \varphi(x) \frac{\partial}{\partial t} m(x, t)dx$$

By definition of an adjoint, then $\int \mathcal{A}\varphi(x)m(x, t)dx = \int \varphi(x)\mathcal{A}^*m(x, t)dx$

since $\int \varphi(x) \frac{\partial}{\partial t} m(x, t)dx = \int \varphi(x)\mathcal{A}^*m(x, t)dx$ holds for every regular φ , the density must be a (weak) solution to

$$\frac{\partial}{\partial t} m(x, t) = \mathcal{A}^*m(x, t)$$

For the process $\hat{x}(t)$ defined by a control $\hat{\nu}(t)$, this gives the Fokker-Planck equation:

$$\frac{\partial m}{\partial t} + A^*m(x, t) + \text{div}(g(x, m, \hat{\nu}(x))m) = 0$$

$$m(x, 0) = m_0(x) \text{ where } A^*m = - \sum_{k,l=1}^n \frac{\partial^2}{\partial x_k \partial x_l} (a_{kl}(x)m(x))$$

Optimization - Hamilton Jacobi Bellman equation

One way to characterize a solution of this optimization problem, given a process $m(t)$, is by a partial differential equation known as the HJB equation which characterizes the value as a function of the state, which must satisfy the PDE if optimal.

Let $u(x, t)$ be the value to the agent of being in state x at time t .

The Lagrangian $L := f(x, m, \nu) + q \cdot g(x, m, \nu)$ represents the instantaneous reward to a decision ν given the state and *costate*, or Lagrange multiplier q on the constraint.

The optimized value of the Lagrangian is the Hamiltonian

$$H(x, m, q) = \inf_{\nu} L(x, m, \nu, q) = L(x, m, \hat{\nu}(x, m, q), q)$$

if this infimum is attained by a function $\hat{\nu}(x, m, q)$

The HJB equation takes the form

$$-\frac{\partial u}{\partial t} + Au = H(x, m, Du)$$

$$u(x, T) = h(x, m(T))$$

Heuristic derivation: start in discrete time and take limit:

The Bellman optimality principle gives:

$$u(x, t) = \inf_{\nu} \{f(x, m, \nu)\Delta t + E_t[u(x(t + \Delta t), t + \Delta t)]\} \quad \forall t \in \{i\Delta t, i \in 0 \dots \frac{T}{\Delta t}\}$$

with terminal condition $u(x(T), T) = h(x(T), m(T))$

Rearrange and divide by Δt

$$0 = \inf_{\nu} \left\{ f(x, m, \nu) + \frac{E_t[u(x(t+\Delta t), t+\Delta t)] - u(x, t)}{\Delta t} \right\}$$

Take limits $\Delta t \rightarrow 0$ and recognize the infinitesimal generator

$$0 = \inf_{\nu} \{f(x, m, \nu) + \mathcal{A}_t u(x, t)\}$$

Expand using the definition

$$0 = \inf_{\nu} \left\{ f(x, m, \nu) + \frac{\partial}{\partial t} u(x, t) + Du(x, t)g(x, m, \nu) + \text{tr}[a(x)D^2u(x, t)] \right\}$$

Move to the LHS anything not depending on ν

$$-\frac{\partial}{\partial t}u(x, t) + Au = \inf_{\nu} \{f(x, m, \nu) + Du(x, t)g(x, m, \nu)\} = H(x, m, Du)$$

The terminal condition is unchanged.

Equilibrium - Fixed point concept

Put everything together, we get the coupled system of PDEs

$$-\frac{\partial u}{\partial t} + Au = \inf_{\nu} [f(x, m, \nu) + Du \cdot g(x, m, \nu)]$$

$$\frac{\partial m}{\partial t} + A^*m(x, t) + \operatorname{div}(g(x, m, \hat{\nu}(x))m) = 0$$

$$m(x, 0) = m_0(x)$$

$$u(x, T) = h(x, m(T))$$

A solution to these is a solution of the MFG.

The rest of the group we will be on applications, extensions, and numerical methods for this problem class.